Advanced Programming Practice · 21CSC203P

Unit 5 · Session 1 · SLO 2 — Solved Worksheet

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Unit â€“ 5 : Session â€“ 1 : SLO â€“ 2 | Modelling Simple Regular Expressions Using Automata

I. Questions & Answers

1. Regular expression for strings ending with 'ab' â€” (a + b)*ab i.e. any string over {a, b} followed by the suffix ab.

2. DFA for a* â€” a single state q0 that is both the start and the accepting state, with a self-loop on 'a'. The empty string is accepted immediately, and every additional 'a' loops back to q0.

3. Converting a regular expression to an NFA â€” use Thompson's construction: build small NFAs for the basic symbols, then compose them with Îµ-transitions for union (|), concatenation, and Kleene star (*), yielding an Îµ-NFA equivalent to the expression.